Project:	Auction 6
Team No.:	6
Class:	CSE 3310; Spring 2026
Module:	Test Plan
Deliverable:	Test Plan Document

Version: [1.2] **Date: [04/11/2026]**

Prepared by:
Team 6

Team members are:

Bruno Garbuio De Souza

Loc Nguyen

Minh Trac

Revision History:

Version number	***Date***	***Originator***	***Reason for change***	***High level description of changes***
1.0	2/16/2026	Team 6	UML Formatting	Formatting and creation of UML Document
1.1	2/27/2026	Team 6	SRA Format	Initiation of the SRA Document
1.2	4/11/2026	Team 6	Test Cases	Implementation of Test Cases

Table of Contents

Team members are: *2*

Revision History: 2

1.Introduction and Project Overview *4*

1.1.Objectives 5

1.1.1 BUSINESS OBJECTIVES 5

1.1.2 SYSTEM OBJECTIVES 5

2.Test Cases *5*

2.1 Registration 5

2.2 Email Verification 5

2.3 Login 6

2.4 Listing Creation 6

2.5 Listing Management 7

2.6. Bid Placement and Validation 7

2.7. Order Processing and Checkout 8

2.8. Shipping State Tracking 8

2.9. Category Browsing 9

2.10. Buy Now 9

3.Project Context Diagram *10*

4. UML System Modeling *11*

4.1 UML-ClassDiagram-System 11

4.2 UML-STD-Login&Registration 11

4.3 UML-ActivityDiagram-Browse&Search 12

4.4 UML-SequenceDiagram-PlaceBid 13

4.5 UML-STD-ListingLifecycle 14

4.6 UML-SequenceDiagram-Checkout&Payment 15

4.7 UML-STD-OrderShipping 16

4.8 UML-ActivityDiagram-AdminModeration 17

5. Systems Requirements *18*

5.1. “Login” and “User” Management Requirements 18

5.2 “Listing” Management Requirements 19

5.3 “Bid” Placement and Validation Requirements 20

5.4 “Order” Processing Requirements **21**

5.5 “Category” Browse Requirements **22**

6. Software Processes and Infrastructure ***23***

Hardware and Infrastructure 23

Hardware Environment 23

Software Environment 23

Data Storage 23

System Components 23

7. Assumptions and Constraints ***24***

7.1. Assumptions **24**

7.2. Constraints **24**

7.3. Out of Scope material **24**

8. Delivery and Schedule ***26***

9. Stakeholder Approval Form ***27***

Appendix: ***28***

1. Introduction and Project Overview

Auction 6 is an Android-based English-style auction car parts marketplace application. The system serves individual buyers and sellers by providing a secure, peer-to-peer on mobile environment. The primary problem the system solves is enabling the community to determine the fair market value of items through competitive, real-time bidding rather than relying on fixed retail pricing, which has proved that sometimes doesnt meet the customers in seller and buyers role. Overall scope of the project focuses on user authentication, listing creation, bidding logic, and simulated transaction processing.

The system centers on fluid user roles, allowing a single authenticated account to act as both a Buyer and a Seller. As a Seller, users can create item listings with images, set starting prices, and optionally set a "Buy Now" price. As a Buyer, users can browse categorized listings, monitor live auctions, and place bids. Additionally, an Administrator role operates in the background to monitor listing content and user activity, ensuring marketplace integrity.

The functional boundaries of Auction 6 include secure user authentication, timed auction management, bid validation, and structured historical tracking. The system explicitly excludes live payment processing and live shipping tracking, opting instead for simulated payment state handling and manual shipping state updates.

Major System Features

- **Secure User Verification**: The system requires users to complete mandatory email verification before placing bids or creating listings to ensure platform safety.
- **English Auction Logic**: The application enforces a strict bidding rule where each new bid must be a minimum of 5% higher than the current highest bid.
- **Timed Listing Management**: Sellers can create item listings with a fixed countdown timer, determining the exact end time of the auction.
- **Historical Data Tracking**: The system generates structured, table-formatted dashboards to display bid histories and user transaction records.
- **Simulated Order Processing**: Upon auction completion, the system determines the winner based on reserve prices and processes simulated payments to finalize orders.

1.1.Objectives

1.1.1 BUSINESS OBJECTIVES

- Achieve a 95% successful completion rate for all auctions where the seller's reserve price is met.
- Maintain a fluid user base where at least 30% of registered accounts utilize both the Buyer and Seller roles within a 90-day period.

1.1.2 SYSTEM OBJECTIVES

- **Performance**: The system shall validate all new bids against the 5% increment rule and update the highest bid amount in under 2 seconds to ensure real-time accuracy.
- **Security**: The system shall restrict bidding and listing creation privileges exclusively to accounts that have successfully passed the email verification process.
- **Reliability**: The system shall execute an automatic order cancellation and a 2-day relisting protocol if simulated payment processing fails after maximum retries.

2.Test Cases

2.1 Registration

Project Name: Auction 6
Test Case Name: Registration
Test Case Id: CSE3310/Fall 2026/Team6/Registration

Test Case No.	Test Case Description	Expected Results	Outcome
TC1	Enter a new valid email and a strong password, then select Register.	Account is created, a verification email is sent, and the user is informed that email verification is required before bidding or listing creation.	
TC2	Enter an email that already exists in the system and attempt to register.	Registration is rejected and a message states that the email is already in use. The form remains visible for correction.	
TC3	Leave one or more required fields blank and select Register.	Registration is blocked and the missing required field or fields are identified to the user.	

2.2 Email Verification

Project Name: Auction 6
Test Case Name: Email Verification
Test Case Id: CSE3310/Fall 2026/Team6/EmailVerification

Test Case No.	Test Case Description	Expected Results	Outcome

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Register successfully and verified with 6-digit code authentication.	Account status changes to verified and the user gains access to bidding and listing creation permissions.	
TC1B	Attempt to place a bid or create a listing before completing email verification.	Action is blocked and a message states that email verification must be completed first.	
TC2	Attempt to verify an account with an incorrect verification code or skip this step	Verification fails and the user is informed to retry the code	
TC3	Navigate back to login from the Verify screen	Back to Login button works	

2.3 Login

Project Name: Auction 6
Test Case Name: Login
Test Case Id: CSE3310/Fall 2026/Team6/Login

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Enter valid registered credentials and select Login.	User is authenticated successfully and redirected to the main marketplace dashboard.	
TC1B	Enter a valid email with an incorrect password and attempt to log in.	Login fails and an error message indicates that the password is incorrect.	
TC2	Enter an email that does not exist in the system and attempt to log in.	Login fails and an error message indicates that the account does not exist or the credentials are invalid.	
TC3	Leave email and or password blank and select Login.	Login is blocked and the user is prompted to complete all required fields.	

2.4 Listing Creation

Project Name: Auction 6
Test Case Name: Listing Creation
Test Case Id: CSE3310/Fall 2026/Team6/ListingCreation

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Log in as an authorized seller, enter valid title, description, starting price, reserve price, auction end time, optional Buy Now price, optionally provide an image URL, and submit.	Listing is created successfully, marked active, and becomes visible to buyers in the selected category.	
TC1B	Create a listing using all required valid fields but without a Buy Now price.	Listing is created successfully and remains available for bidding without a Buy Now option.	
TC2	Leave one or more required fields blank and attempt to submit the listing.	Listing creation fails and the user is prompted to complete all required fields.	
TC3	Enter an auction end time in the past and attempt to submit the listing.	Listing creation fails and an error indicates that the auction end time must be in the future.	

Test Case No.	Test Case Description	Expected Results	Outcome
TC4	Attempt to create a listing while logged in without seller authorization.	Listing creation is blocked and the user is informed that seller privileges are required.	

2.5 Listing Management

Project Name: Auction 6
Test Case Name: Listing Management
Test Case Id: CSE3310/Fall 2026/Team6/ListingManagement

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Open a seller-owned active listing from the seller listings screen.	Listing details load correctly, including title, description, images, prices, end time, and active status.	
TC1B	Delete or cancel a seller-owned listing before any business rule blocks the action.	Listing is removed or marked inactive according to system rules and is no longer visible to buyers.	
TC2	Allow a listing to reach its scheduled end time.	The listing automatically closes and no additional bids are accepted after close.	
TC3	View the minimum next acceptable bid on an active listing.	The system displays the correct minimum next bid based on the current highest bid and the global 5 percent increment rule.	

2.6. Bid Placement and Validation

Project Name: Auction 6
Test Case Name: Bid Placement and Validation
Test Case Id: CSE3310/Fall 2026/Team6/BidPlacementValidation

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Open an active listing as an authorized buyer and place a bid that is more than 5 percent above the current highest bid before the end time.	Bid is accepted, recorded in bid history, and the listing current highest bid is updated.	
TC1B	Place a bid that is exactly equal to the minimum acceptable next bid.	Bid is accepted and becomes the new highest bid for the listing.	
TC2	Place a bid below the current highest bid.	Bid is rejected and an error states that the bid must be higher than the current highest bid.	
TC3	Place a bid higher than the current bid but below the 5 percent minimum increment.	Bid is rejected and the required minimum next bid is shown to the user.	
TC4	Attempt to place a bid after the auction end time has passed.	Bid is rejected and the listing remains closed.	
TC5	Attempt to place a bid on an inactive or closed listing.	Bid is rejected and an error indicates that the auction is no longer active.	

Test Case No.	Test Case Description	Expected Results	Outcome
TC6	Attempt to place a bid using an unverified account.	Bid is rejected and the user is informed that email verification is required.	
TC7	Two buyers place valid bids in sequence on the same listing.	The bids are recorded in timestamp order and the highest valid bid is displayed correctly.	

2.7. Order Processing and Checkout

Project Name: Auction 6
Test Case Name: Order Processing and Checkout
Test Case Id: CSE3310/Fall 2026/Team6/OrderProcessingCheckout

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Allow an auction to end with a highest bid that meets or exceeds the reserve price.	The system automatically generates an order for the winning bidder and records the final transaction price.	
TC1B	Allow an auction to end when the highest bid does not meet the reserve price.	No winning order is generated and the item remains unsold or follows relisting rules.	
TC2	Select the Buy Now option on an active listing.	The auction closes immediately for that buyer and an order is generated at the Buy Now price.	
TC3	Enter a valid shipping address and authorize simulated payment successfully.	Payment status updates to Authorized, the seller is notified to ship the item, and the order remains active.	
TC4	Simulate a payment failure during checkout.	Payment status updates to Failed and the system follows failure handling logic, including cancellation or relisting if applicable.	
TC5	Cancel an order during an allowed cancellation flow after payment failure.	Order status is updated appropriately and the listing becomes eligible for relisting according to system rules.	
TC6	View order details after a successful auction close.	Order details show buyer, seller, final price, payment status, and shipping status correctly.	

2.8. Shipping State Tracking

Project Name: Auction 6
Test Case Name: Shipping State Tracking
Test Case Id: CSE3310/Fall 2026/Team6/ShippingStateTracking

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	After successful order creation, the seller updates shipping state from Not Shipped to Label Created.	Shipping status is updated and the new state is visible to the buyer in the order record.	
TC1B	Seller updates shipping state from Label Created to In Transit.	The new shipping state is saved and shown in the order history.	

Test Case No.	Test Case Description	Expected Results	Outcome
TC2	Seller updates shipping state to Delivered and the buyer confirms delivery.	The order is marked delivered or complete according to implementation rules.	
TC3	Mark an order as Returned.	Order record updates to Returned and remains visible in transaction history.	

2.9. Category Browsing

Project Name: Auction 6
Test Case Name: Category Browsing
Test Case Id: CSE3310/Fall 2026/Team6/CategoryBrowsing

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Open category browsing and select a category that contains active listings.	The system displays all active listings assigned to the selected category.	
TC1B	Select a category that contains no active listings.	The system shows an empty-state message indicating that no listings are available in that category.	
TC2	Switch from one category to another.	The listing display refreshes correctly and only listings for the newly selected category are shown.	
TC3	Retrieve listings from a category after one listing in that category has closed.	The closed listing is no longer shown among active listings for that category.	

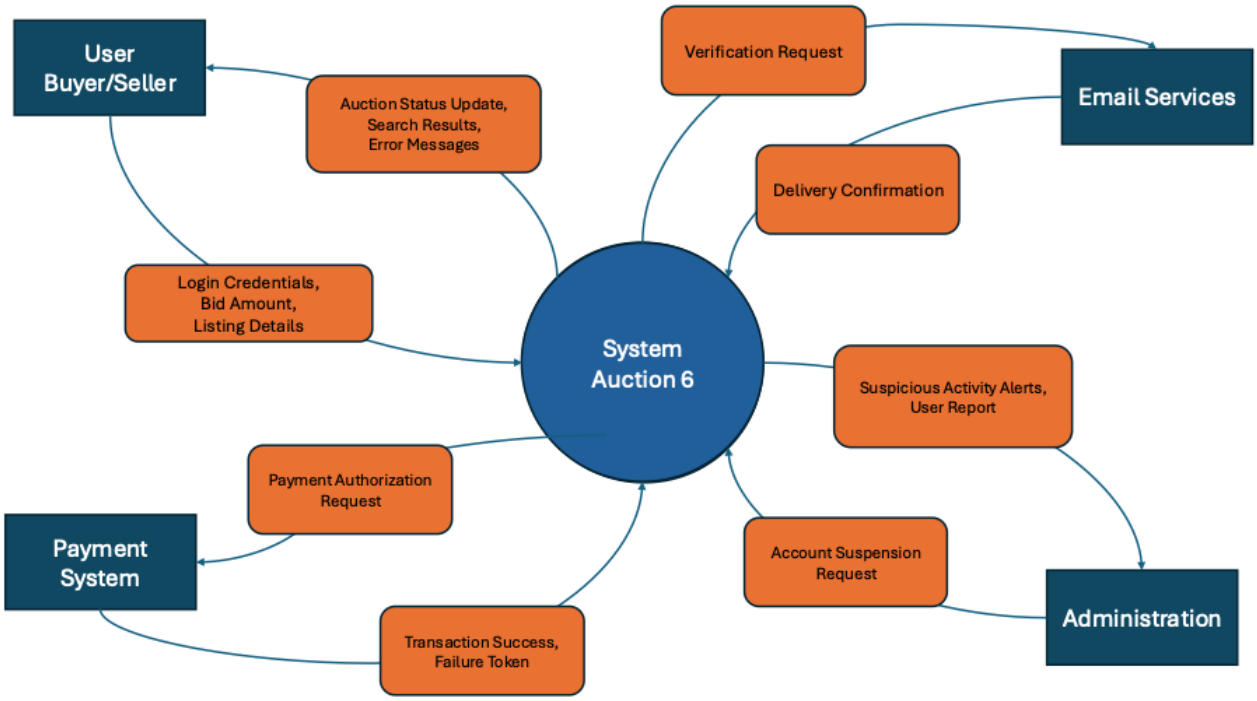
2.10. Buy Now

Project Name: Auction 6
Test Case Name: Buy Now
Test Case Id: CSE3310/Fall 2026/Team6/BuyNow

Test Case No.	Test Case Description	Expected Results	Outcome
TC1A	Open a listing with a Buy Now price and select Buy Now before the auction closes.	The system creates an order immediately at the Buy Now price and prevents any further bidding on that listing.	
TC1B	Open a listing that does not have a Buy Now price.	The Buy Now action is hidden or disabled for that listing.	
TC2	Attempt to use Buy Now on a closed listing.	The action is rejected and the user is informed that the listing is no longer active.	
TC3	Attempt to use Buy Now with an unverified account.	The action is rejected and the user is informed that email verification is required.	

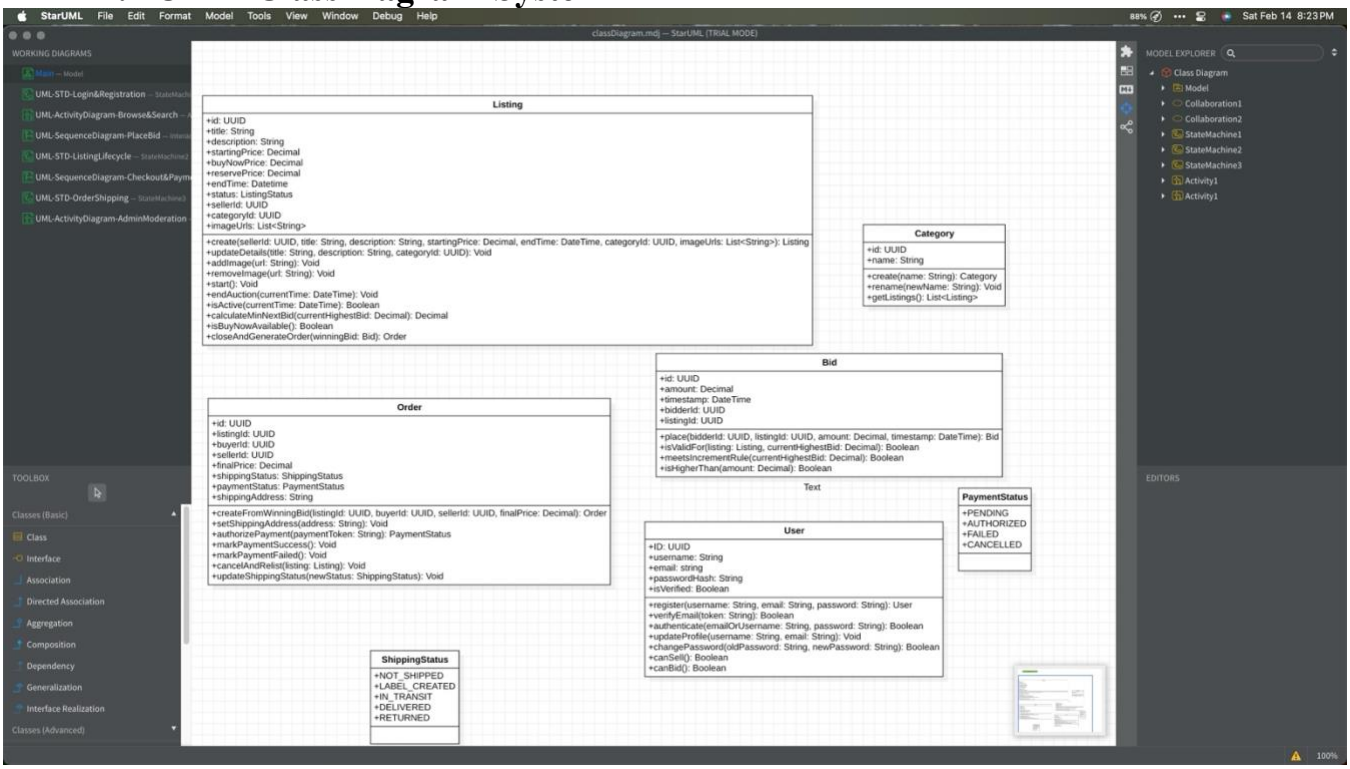
3.Project Context Diagram

The following is a context diagram displaying all high-level components of the system.

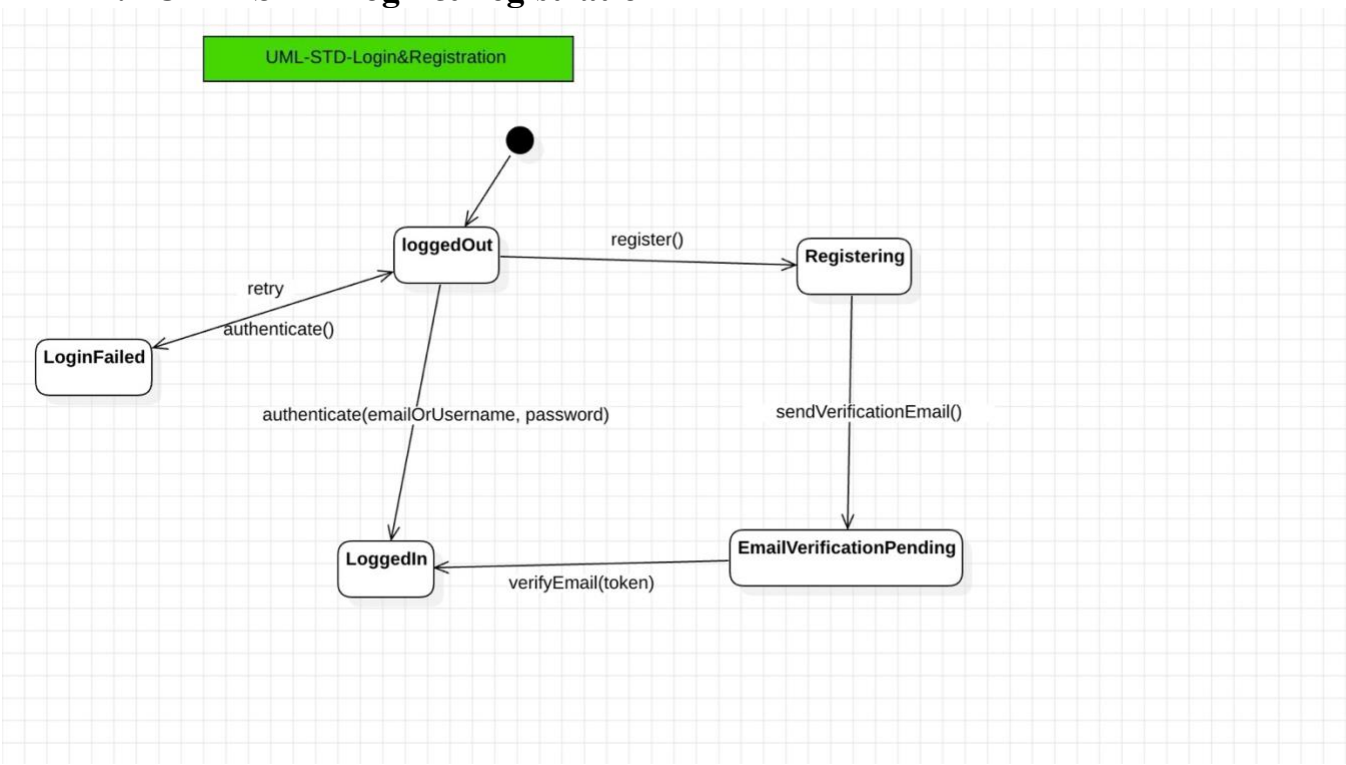


4. UML System Modeling

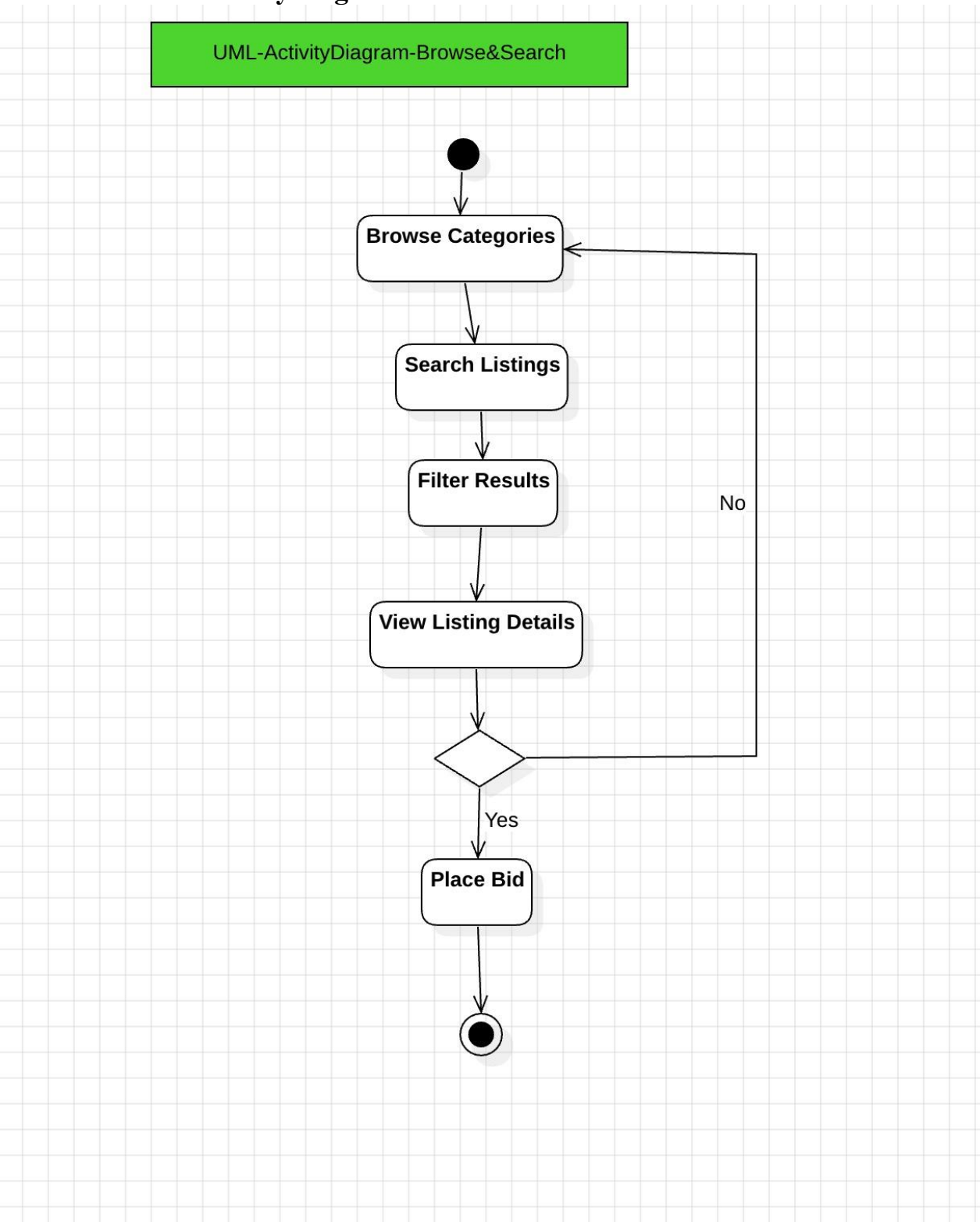
4.1 UML-ClassDiagram-System



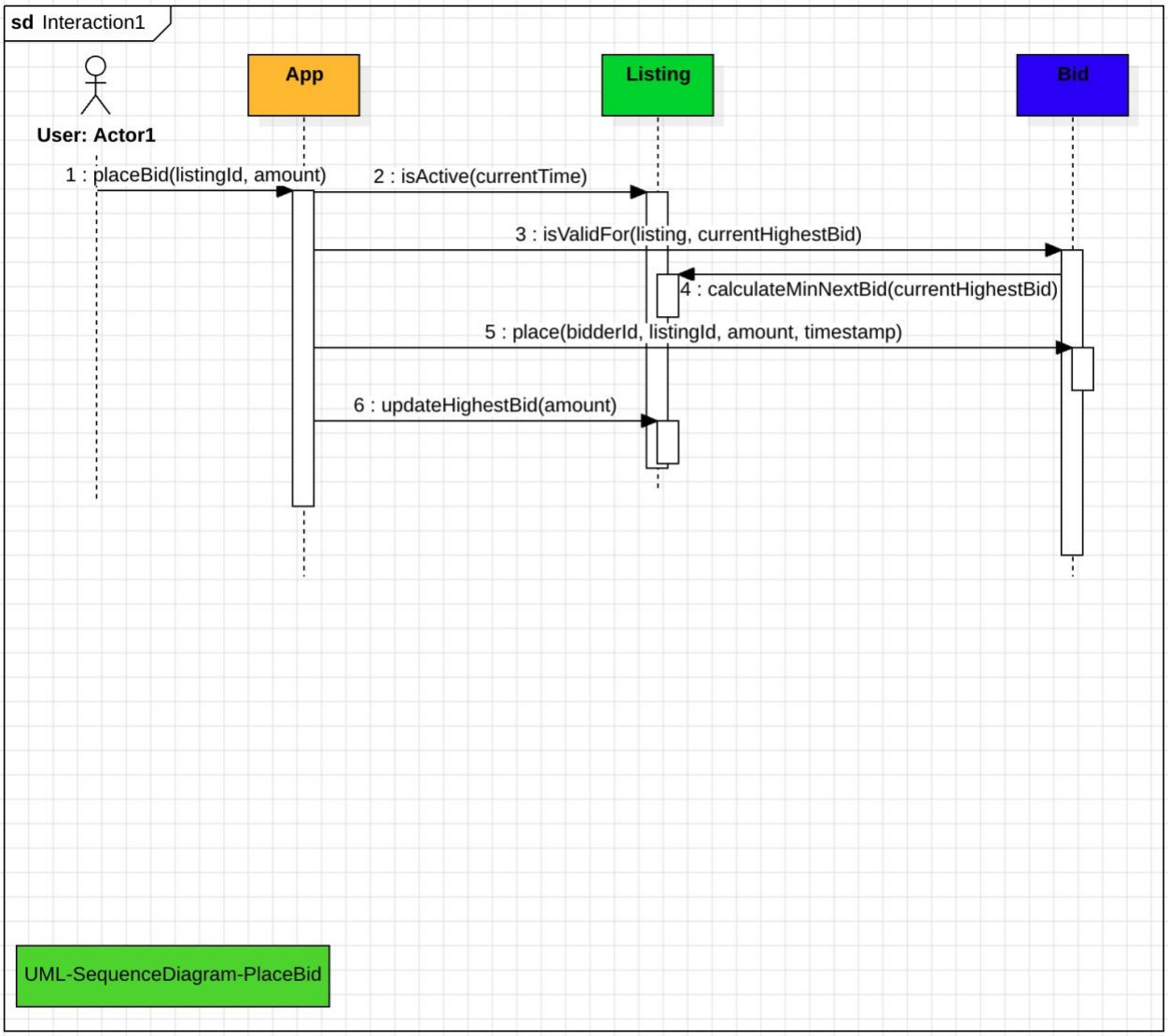
4.2 UML-STD-Login&Registration



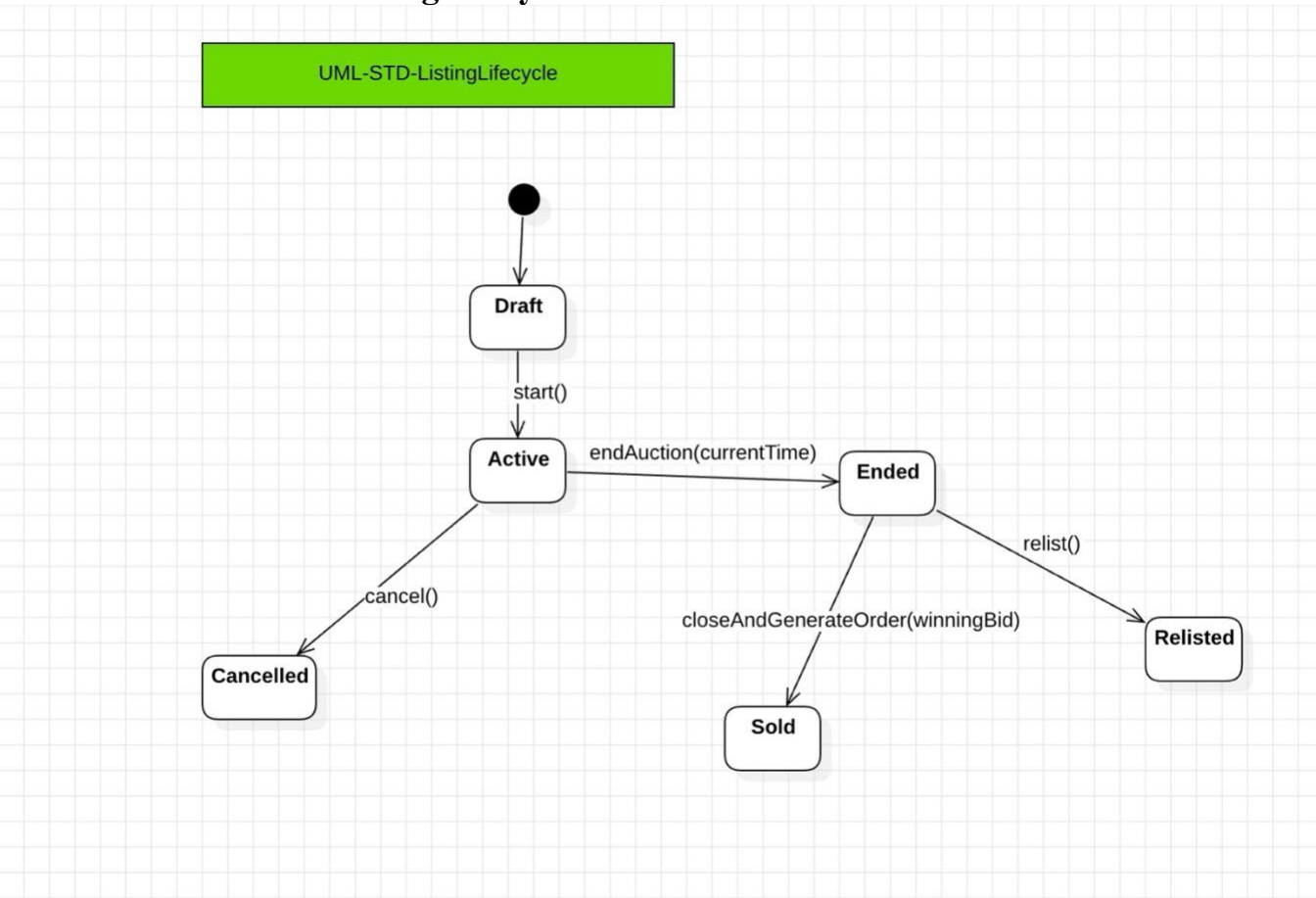
4.3 UML-ActivityDiagram-Browse&Search



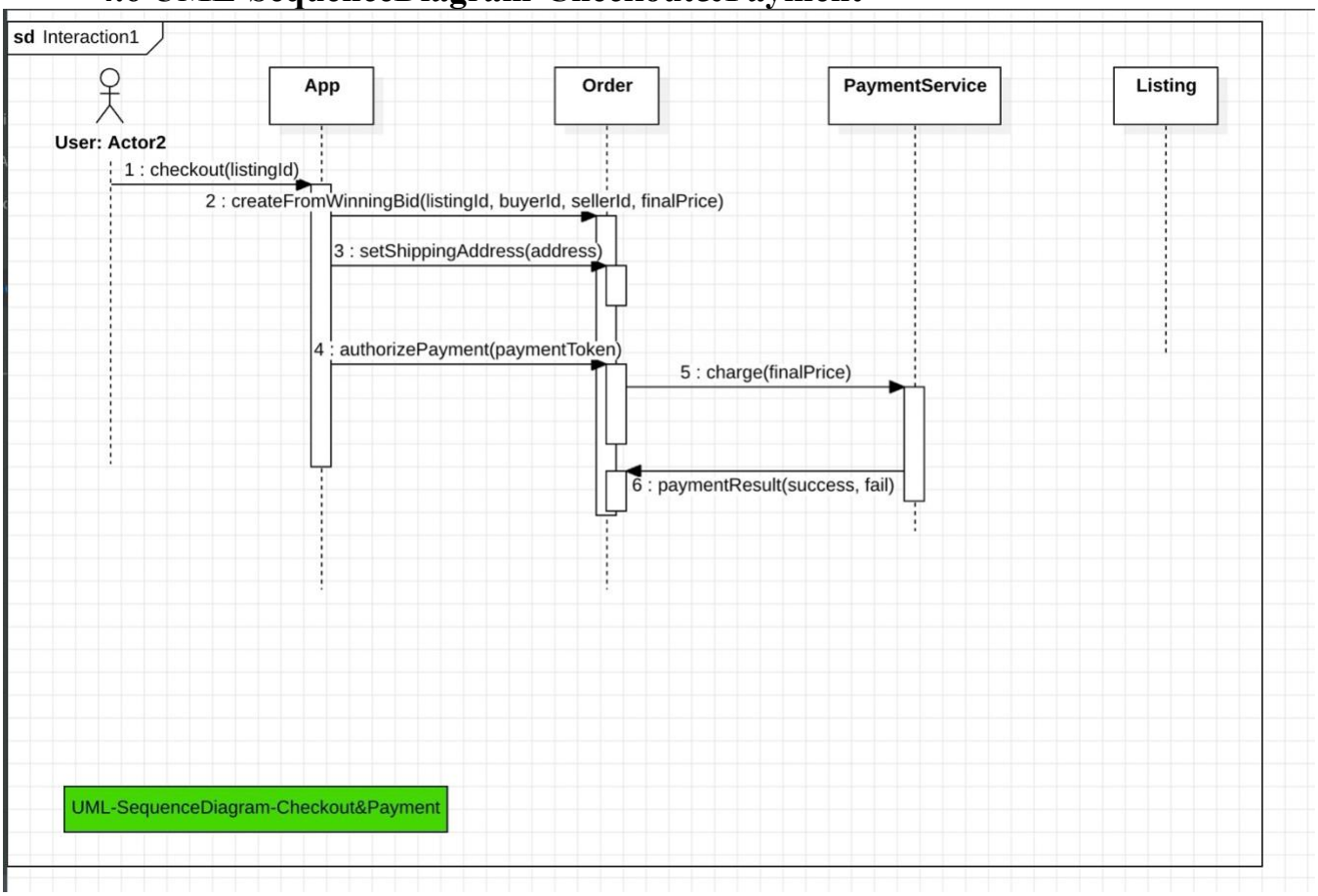
4.4 UML-SequenceDiagram-PlaceBid



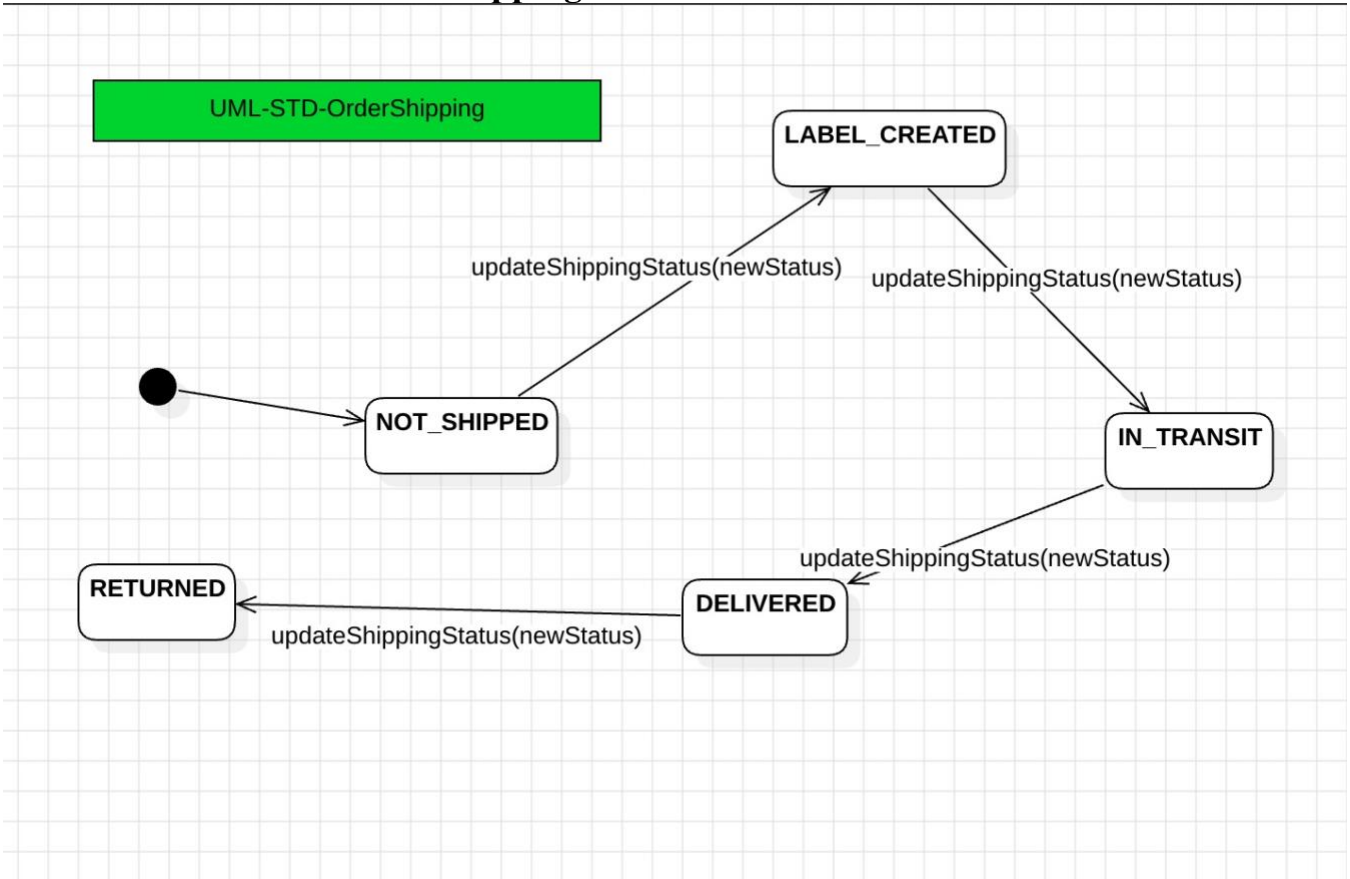
4.5 UML-STD-ListingLifecycle



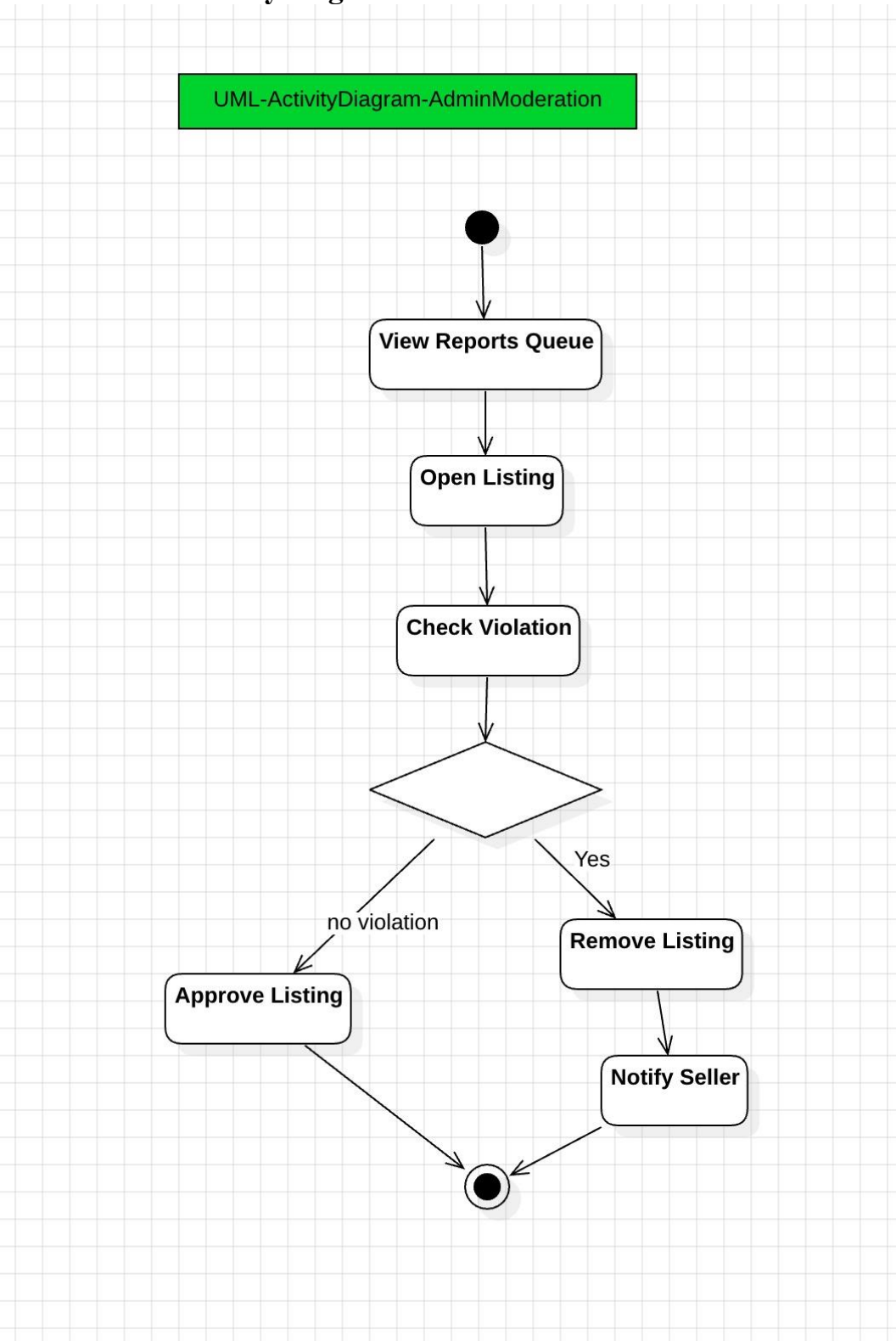
4.6 UML-SequenceDiagram-Checkout&Payment



4.7 UML-STD-OrderShipping



4.8 UML-ActivityDiagram-AdminModeration



5. Systems Requirements

The following section breaks down each objective by describing the requirements needed in order to fully understand the functionality of the system. This focuses on the user functionality and does not include information regarding the hardware, database, test data, or android versions.

5.1. “Login” and “User” Management Requirements

	User management
Requirement Title:	User Account Management
Sequence No:	1.0
Short description:	Handles user registration, authentication, and profile management.
Description:	The system shall allow users to register for a new account using an email and password, and verify their email address. Users must be able to securely log in (authenticate), update their profile information, and change their passwords. The system will also determine if the user has the appropriate permissions to buy (canBid) or sell (canSell).
Pre-Conditions:	The user has downloaded the application.
Post Conditions:	A valid user account is created, authenticated, and updated in the system, granting the user access to buyer or seller features.
Other attributes:	N/A

5.2 “Listing” Management Requirements

	Listing Management
Requirement Title:	Auction Listing Management
Sequence No:	2.0
Short description:	Enables sellers to create, manage, and monitor auction items.
Description:	The system shall allow authorized sellers to create a new item listing by providing a title, description, starting price, optional 'Buy Now' price, reserve price, and auction end time. Sellers can add or remove images from the listing. The system will track whether the auction is currently active, calculate the minimum next acceptable bid, and automatically close the auction to generate an order once the end time is reached.
Pre-Conditions:	The user must be logged in and authorized with seller privileges.
Post Conditions:	A listing becomes active and visible to buyers. Upon completion, the listing is closed, and if a valid bid was placed, an order is generated.
Other attributes:	N/A

5.3 “Bid” Placement and Validation Requirements

	Bid Management
Requirement Title:	Bid Placement and Validation
Sequence No:	3.0
Short description:	Allows buyers to securely place bids on active auction listings.
Description:	The system shall allow users to place bids on active listings. Before accepting a bid, the system must validate that the bid amount is higher than the current highest bid, meets the minimum bid increment rule, and is placed before the auction's scheduled end time.
Pre-Conditions:	The user must be logged in, authorized to bid, and viewing an active listing.
Post Conditions:	The bid is successfully recorded, and the listing's current highest bid is updated.
Other attributes:	N/A

5.4 “Order” Processing Requirements

	Order Processing
Requirement Title:	Order Processing and Checkout
Sequence No:	4.0
Short description:	Manages the transaction and fulfillment process after an auction is won.
Description:	The system shall automatically generate an order when an auction closes with a winning bid or when a user selects a 'Buy Now' option. The user can set a shipping address and authorize payment. The system will track the payment status (Pending, Authorized, Failed, Cancelled) and the shipping status (Not Shipped, Label Created, In Transit, Delivered, Returned). If a payment fails or an order is cancelled, the system shall allow the listing to be relisted.
Pre-Conditions:	An auction has ended with a winning bid, or a buyer has triggered a 'Buy Now' purchase.
Post Conditions:	Payment is processed, the seller is notified to ship the item, and the shipping status is tracked until delivery.
Other attributes:	N/A

5.5 “Category” Browse Requirements

	Item Categories
Requirement Title:	Item Categorization
Sequence No:	5.0
Short description:	Organizes listings into distinct categories for browsing.
Description:	The system shall allow listings to be grouped into specific categories to facilitate user browsing and searching. The system must support the creation and renaming of categories and allow users to retrieve all active listings associated with a specific category.
Pre-Conditions:	The system is operational, and users are attempting to browse listings.
Post Conditions:	Users are presented with filtered lists of items based on their selected category.
Other attributes:	N/A

6. Software Processes and Infrastructure

Hardware and Infrastructure

The Auction 6 system will operate within an Android mobile environment and will utilize modern Android development tools and frameworks to ensure compatibility, performance, and maintainability.

Hardware Environment

- **Development Platform:** MacOS or Windows development workstation capable of running Android Studio
- **Testing Device: Android** Emulator (Pixel 8) or equivalent physical Android devices.
- **Minimum Supported Android Version**: Android 11

Software Environment

- **Development IDE:** Android Studio (Panda or newer version)
- **Programming Language**: Kotlin
- **UI Framework:** Jetpack Compose
- **Architecture Pattern:** MVVM **(Model-View-ViewModel).**
- **Version Control:** Git and GitHub repository for source code management.

Data Storage

- For the Minimum Viable Product (MVP), the system will use:
  Database: SQLite
- The database will store:
  User account information
  Auction listing
  Bid history
  Order record
  Item categories

System Components

The application consists of the following major subsystems:

1. **Authentication Module**
   - Handle login, registration, and user verification.
2. **Listing Management Module**
   - Allows sellers to create and manage auction listings.
3. **Bid Management Module**
   - Validates and records bids.
4. **Order Processing Module**
   - Handles simulated order completion and payment state tracking.
5. **Category Browsing Module**
   - Allows users to filter and browse listings by category.

7. Assumptions and Constraints

7.1. Assumptions

The following assumptions were made during system design and development planning:

- Users possess a smartphone capable of running Android 11 or newer.
- Users have access to a stable internet connection for real-time bidding interactions.
- All users will complete the email verification process before participating in auctions.
- Sellers provide accurate and truthful item descriptions.
- Auction bids are processed sequentially, and timestamps are reliable for determining bid order.

7.2. Constraints

The system operates under the following constraints:

- **Platform Constraints:**
  The application must run exclusively on Android devices.

- **Database Constraints:**
  The MVP will use **SQLite** for local data storage rather than a cloud-based database.

- **Development Timeline Constraint:**
  The project must be completed within the academic semester timelinefor CSE 3310.

- **Security Constraint:**
  Authentication must enforce secure login credentials and email verification before users can access auction features.

- **Scope Constraint:**
  The system will simulate payment processing rather than integrating real payment gateways.

7.3. Out of Scope material

The follow features are explicitly excluded from the MVP version of auction 6:

- Real payment gateway integration (PayPal, Stripe, etc.)
- Real shipping carrier tracking integration
- Fraud detection or automated moderation systems
- Web browser version of the marketplace
- Real-time push notifications
- AI-based price recommendation systems.

These features may be considered for future versions but are not included in the current project scope.

8. Delivery and Schedule

Task/Milestone Description	Anticipated Start Date	Anticipated End Date	Status	Comments
Project Introduction & Environment Setup (Android Studio Installation)	Jan 12, 2026	Jan 21, 2026	Completed	Android Studio installed and development environment configured
Project Description	Jan 21, 2026	Jan 23, 2026	Completed	One-page project description submitted
UML Design & System Modeling (Increment 1)	Jan 28, 2026	Feb 13, 2026	Completed	Use case diagram, class diagram, and system model completed
Android App Functional Setup (Login / UI / DB Connection)	Feb 16, 2026	Feb 23, 2026	Completed	Android app configured with login and database connection
SRA Document Development (Increment 2)	Feb 18, 2026	Mar 02, 2026	Completed	System Requirements Analysis document prepared
Test Plan Development (Increment 3)	Feb 25, 2026	Apr 11, 2026	In Progress	Test plan documentation and testing strategy
Security & Architecture Improvements	Mar 16, 2026	Apr 24, 2026	In Progress	Improvements based on system architecture and security lectures
Agile Development & Feature Finalization	Apr 06, 2026	Apr 13, 2026	In Progress	Final system functionality and improvements
Project Presentation Preparation	Apr 13, 2026	Apr 20, 2026	In Progress	Prepare demo, slides, and final documentation
Final Milestone: Project Delivery (Increment 4)	Apr 20, 2026	Apr 24, 2026	In Progress	Final submission including project binder and evaluation forms

9. Stakeholder Approval Form

Stakeholder Name	Stakeholder Role	Stakeholder Comments	Stakeholder Approval Signature and Date
Bhanu Jain	Project Manager		
Shaikh Islam	Project Assistant		
Aman Gulati	Project Assistant		
Bruno Garbuio De Souza	Developer		
Loc Nguyen	Developer		
Minh Trac	Developer		

Appendix:

Appendix A: Glossary of Terms

- **English Auction**: An open-outcry ascending price auction where participants bid against each other, with each subsequent bid required to be higher than the previous one.
- **Reserve Price**: A hidden minimum price set by the seller; if the highest bid at the end of the auction does not meet this threshold, the item remains unsold.
- **MVP (Minimum Viable Product)**: The version of the application containing only the core functional pillars (authentication, bidding, listing, and simulated checkout) required for initial deployment and testing.
- **Fluid Roles**: An architectural design where a single authenticated user account possesses the permissions to act as both a buyer and a seller without requiring separate accounts.